Loveland, Morrissey & Seaborg

Modern Nuclear Chemistry (2nd ed., 2017)

Chapter 8 — Beta Decay
Sections 8.1–8.2: Introduction & Neutrino Hypothesis
Refresher Notes

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Summer 2026

These notes are paced as a slow-burn refresher. Section 0 re-establishes prerequisite vocabulary you may be rusty on after a few months away, then we step through the physics of §8.1 and §8.2 with deliberate attention to motivation and to the small algebraic moves that are easy to gloss over.

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0. Background Refresher (prerequisites)

Before opening Ch. 8, it helps to have a few earlier ideas fresh in your head. Beta decay sits on top of all of them.

0.1 Nuclear notation and basic kinematics

A nucleus is written

$$^{A}_{Z}\mathrm{X}_{N}, \qquad A = Z + N,$$

where Z is the proton (atomic) number, N the neutron number, and A the mass number. *Isobars* share A but differ in Z — the entire game of beta decay is hopping between isobars.

The atomic mass excess is defined as

$$\Delta(^{A}\mathrm{X}) \;\equiv\; \left[M(^{A}\mathrm{X}) - A\,\mathrm{u}\right] c^2,$$

with $1\,\mathrm{u} \approx 931.494\,\mathrm{MeV}/c^2$. Mass excesses are tabulated (e.g., AME2020) because differences of mass excesses give Q-values directly:

$$Q = \sum_{\text{initial}} \Delta - \sum_{\text{final}} \Delta.$$

This is the single most useful identity in radioactive-decay bookkeeping.

0.2 Q-value sign convention

A reaction or decay is *exoergic* (it can occur spontaneously, energetically) when $Q > 0$. The released energy appears as kinetic energy of the products:

$$Q = T_{\text{final}} - T_{\text{initial}} = \left(M_i - M_f\right) c^2.$$

For beta decay, the reactant nucleus is at rest in the lab to good approximation, so $T_{\text{initial}} \approx 0$ and Q is the total kinetic energy shared by the products.

0.3 Conservation laws you must always invoke

Every nuclear-physics calculation enforces, in order:

- **Energy** (mass–energy, including rest masses).
- **Linear momentum** (3-vector).
- **Electric charge** Q_{em}.
- **Baryon number** B (protons and neutrons each carry $B = +1$).
- **Lepton number** L (electron $L_e = +1$, positron -1, ν_e $+1$, $\bar{\nu}_e$ -1).
- **Angular momentum** (and, for allowed transitions, *parity*).

You will see, in §8.2, that the historical introduction of the neutrino was forced by the demand to *simultaneously* preserve energy, momentum, and angular momentum.

0.4 The decay-mode landscape (what comes before §8.1)

By the start of Ch. 8 you have already met:

- α decay: ${}^{A}_{Z}\mathrm{X} \to {}^{A-4}_{Z-2}\mathrm{Y} + \alpha$ (Ch. 7).
- Spontaneous fission and cluster decays.
- γ emission and internal conversion (Ch. 9, but already referenced).

Beta decay is the missing piece that connects nuclei of the *same* A — sliding along an isobaric chain toward stability.

Background Refresher

Mental picture. Plot the isobars of fixed A versus Z. The atomic masses lie on a parabola (or two parabolas, for odd A). The nucleus at the bottom of the parabola is β-stable. Everything else funnels toward it via β^-, β^+, or electron capture (EC). §8.1 is the introduction to that funnel; §8.2 explains why the funnel emits *three* particles, not two.

1 Section 8.1 — Introduction to Beta Decay

1.1 Why "beta" decay deserves a long chapter

Beta decay is the most common form of radioactivity. Of the ∼3000 known nuclides, the overwhelming majority decay by β^-, β^+, or EC. Ch. 8 spends so many pages on it because *the underlying physics is the weak interaction itself* — the same interaction that powers solar fusion, sets the lifetime of the free neutron, and produces every neutrino your body has ever encountered. Studying beta decay gives you the cleanest laboratory example of a weak process.

1.2 The three modes

For a parent atom ${}^{A}_{Z}\mathrm{X}$ decaying to a daughter ${}^{A}_{Z'}\mathrm{Y}$, there are three weak channels. Memorize the bookkeeping:

Key Idea

Negatron emission (β^-):

$$ {}^{A}_{Z}\mathrm{X} \;\to\; {}^{A}_{Z+1}\mathrm{Y} + e^- + \bar{\nu}_e \qquad (n \to p + e^- + \bar{\nu}_e). $$

Positron emission (β^+):

$$ {}^{A}_{Z}\mathrm{X} \;\to\; {}^{A}_{Z-1}\mathrm{Y} + e^+ + \nu_e \qquad (p \to n + e^+ + \nu_e). $$

Electron capture (EC):

$$ {}^{A}_{Z}\mathrm{X} + e^-_{\text{atomic}} \;\to\; {}^{A}_{Z-1}\mathrm{Y} + \nu_e \qquad (p + e^- \to n + \nu_e). $$

In all three, A is conserved (no nucleon leaves the nucleus); only Z changes by ± 1. Equivalently, β^- converts a neutron into a proton inside the nucleus, while β^+ and EC do the opposite.

1.3 Q-values written in atomic masses

Loveland gives nuclear-mass formulas, but in practice we use *atomic* masses (which include all Z electrons). The electron-mass bookkeeping is where students slip. Write the parent atomic mass as $M({}^{A}\mathrm{X}) = M_{\text{nuc}}({}^{A}\mathrm{X}) + Z\,m_e$ (binding energy of electrons neglected at the keV-or-better level relevant here), and proceed:

Negatron emission.

$$ Q_{\beta^-}/c^2 = \big[M_{\text{nuc}}(\mathrm{X}) - M_{\text{nuc}}(\mathrm{Y}) - m_e\big]. $$

Convert to atomic masses by adding $Z\,m_e$ to the parent and subtracting $(Z+1)\,m_e$ from the daughter; the leftover m_e's cancel:

$$ \boxed{Q_{\beta^-} = \big[M({}^{A}_{Z}\mathrm{X}) - M({}^{A}_{Z+1}\mathrm{Y})\big]\,c^2.} $$

The neutrino is massless (or near-massless) and drops out cleanly.

Positron emission. The same algebra now requires *adding* the daughter's electron count and *subtracting* the parent's, but you also have to account for the emitted e^+:

$$ \boxed{Q_{\beta^+} = \big[M({}^{A}_{Z}\mathrm{X}) - M({}^{A}_{Z-1}\mathrm{Y}) - 2m_e\big]\,c^2.} $$

The factor of $2m_ec^2 \approx 1.022\,\mathrm{MeV}$ is the *positron-emission threshold*: positron emission cannot occur unless the parent–daughter atomic-mass difference exceeds $1.022\,\mathrm{MeV}$.

Electron capture. Here the captured e^- is one of the parent's own atomic electrons, so the electron-mass cancellation is different:

$$\boxed{Q_{\mathrm{EC}} = \left[M({}^{A}_{Z}\mathrm{X}) - M({}^{A}_{Z-1}\mathrm{Y})\right]c^2 - B_n.}$$

B_n is the binding energy of the captured electron in its atomic shell (K-shell capture is most common; B_K is a few keV to ~ 100 keV depending on Z).

Common Pitfall

Watch the $2m_ec^2$. Whenever a positron appears in a final state, two "extra" electron rest energies have to be subtracted in the atomic-mass formula. Forgetting this is the single most common mistake when classifying which mode is energetically allowed.

1.4 Energy windows: when does each mode operate?

Comparing the three boxed formulas:

- β^- runs whenever $M(\text{parent}) > M(\text{daughter})$.
- EC runs whenever $M(\text{parent}) > M(\text{daughter}) + B_e$ — a very small threshold (typically keV).
- β^+ runs only when $M(\text{parent}) > M(\text{daughter}) + 2m_ec^2$ ($\approx 1.022\,\mathrm{MeV}$).

Key Idea

For proton-rich nuclei, EC and β^+ *compete*. EC always wins at low Q (because it has no $2m_ec^2$ penalty), and the two coexist when $Q_{\beta^+} > 0$. As Z increases, EC dominates because the inner-shell electron density at the nucleus grows rapidly with Z.

1.5 Half-lives and the spectrum of decays

Beta-decay half-lives span more than 25 orders of magnitude — from milliseconds for very neutron-rich exotic nuclei to $> 10^{20}$ yr for double-beta emitters. This enormous range is set by:

- the available phase space (roughly $\propto Q^5$ for allowed decays: the famous *Sargent rule*, derived from Fermi theory in §8.4),
- nuclear matrix elements (selection rules; §8.5–§8.6),
- Coulomb effects on the outgoing charged lepton (the function $F(Z, W)$ that you will meet in the Kurie plot).

Section 8.1 only previews these; §8.4–§8.6 do the heavy lifting.

1.6 Why a chapter, not a paragraph

The opening of Ch. 8 is essentially a contract: it lists the three modes, states the energetic conditions, points to nuclide-chart trends, and sets up the physics puzzle that §8.2 then solves — the missing energy and the missing spin. Understanding the rest of the chapter is a matter of filling in mechanism (§8.3 Fermi theory), sharpening the bookkeeping (§8.4 Kurie plots, ft values), and classifying transitions (§8.5–§8.7). Hold onto the three boxed Q-formulas; they are the arithmetic spine of the chapter.

2 Section 8.2 — The Neutrino Hypothesis

2.1 The crisis of 1914–1930

Here is the historical knot that §8.2 untangles. Around 1914 Chadwick measured the β-particle energy spectrum and made a discovery that was, at the time, alarming:

Key Idea

Two-body decay implies a sharp line; experiment shows a continuum. If beta decay were truly a two-body process

$$\mathrm{X} \to \mathrm{Y} + e^-,$$

energy and momentum conservation would force the electron to come out with *a single, fixed* kinetic energy $T_e = Q$ (up to a tiny nuclear-recoil correction). Experimentally, the electrons emerge with a *continuous* distribution of energies between 0 and a sharp endpoint $T_{\max}$. The endpoint matches the expected Q. The bulk of the spectrum lies below it. *Where does the missing energy go?*

This was deeply unsettling. Bohr was prepared, briefly, to abandon energy conservation in nuclear reactions. Pauli was not.

2.2 Pauli's 1930 letter

In December 1930, Pauli wrote his famous open letter ("Dear Radioactive Ladies and Gentlemen ...") addressed to a workshop in Tübingen, proposing the existence of a new neutral, light, weakly interacting particle emitted *together with* the electron. He called it the *neutron* — unfortunate, since two years later Chadwick discovered the strongly interacting neutron we now know. Fermi later renamed Pauli's particle *neutrino* ("little neutral one") and built the first quantitative theory of beta decay around it (1934, the subject of §8.3).

The neutrino restores three conservation laws at once:

1. **Energy.** A three-body final state shares Q continuously between the electron and the (anti)neutrino, producing a continuous electron spectrum with the correct endpoint.
2. **Momentum.** The unobserved neutrino carries off the "missing" momentum; the daughter recoil and the electron no longer need to be back-to-back.
3. **Angular momentum.** Without the neutrino, the spin bookkeeping is off by $\hbar/2$ in every β decay. With a spin-$\frac{1}{2}$ neutrino, the change of nuclear spin can be $\Delta J = 0$ or 1 in allowed decays — exactly what is observed.

Worked Example

The ^{14}N statistics paradox (paraphrased from the period). ^{14}N has integer spin and obeys *Bose* statistics. If the nucleus contained 14 protons and 7 "intra-nuclear electrons" (the pre-1932 model), the count of fermions would be 21 — predicting Fermi statistics, contrary to experiment. Once the neutron was discovered and β decay reinterpreted as $n \to p + e^- + \bar{\nu}_e$, ^{14}N became $7p + 7n$ (14 fermions, integer total spin, Bose statistics) — and the paradox dissolved.

2.3 What the neutrino must look like

From the conservation arguments alone, one can deduce the neutrino's properties *before* ever detecting it:

- **Charge:** 0 (charge already balances on the lepton side).
- **Spin:** $\frac{1}{2}$ (only way to fix angular-momentum bookkeeping in allowed decays).
- **Mass:** very small. The shape of the β spectrum near its endpoint goes as $(T_{\max} - T)^2$ for $m_\nu = 0$ but bends differently for nonzero mass; pre-1956 data already showed $m_\nu c^2 \ll m_e c^2$. (Modern bounds: $m_{\nu_e} \lesssim 0.8\,\mathrm{eV}$ from KATRIN.)
- **Cross-section:** extraordinarily small — a free $\sim$1 MeV neutrino has a cross-section $\sim 10^{-44}\,\mathrm{cm}^2$, with mean free path of order many light-years of water. This is why direct detection took 26 years.

2.4 Neutrino vs. antineutrino

The neutrino emitted in β^- decay is the *antineutrino* $\bar{\nu}_e$. The one emitted in β^+ and the one absorbed in EC is the *neutrino* ν_e. The bookkeeping that enforces this distinction is *lepton number conservation* ($L_e = +1$ for e^-, ν_e; $L_e = -1$ for $e^+, \bar{\nu}_e$). Check the three modes:

$$\begin{aligned}
\beta^- :&\quad n \to p + e^- + \bar{\nu}_e, \quad 0 \to 0 + 1 + (-1) = 0 \\
\beta^+ :&\quad p \to n + e^+ + \nu_e, \quad 0 \to 0 + (-1) + 1 = 0 \\
\mathrm{EC} :&\quad p + e^- \to n + \nu_e, \quad 0 + 1 \to 0 + 1
\end{aligned}$$

Lepton number balances in every channel.

2.5 Direct detection: Reines and Cowan, 1956

It took until the mid-1950s for Reines and Cowan to detect $\bar{\nu}_e$ directly, using inverse beta decay ($\bar{\nu}_e + p \to n + e^+$) on a target placed near the Savannah River reactor. The signature was a delayed coincidence between the positron annihilation photons and the neutron capture on cadmium. Pauli's hypothesis became Pauli's particle.

2.6 The conceptual punchline of §8.2

Key Idea

β decay is a **three-body** weak process. The electron's continuous spectrum, the apparent angular-momentum mismatch, and the nuclear-statistics paradox all dissolve once you accept that an (anti)neutrino is co-emitted. The shape of the electron spectrum, the endpoint behavior, and ultimately the lifetime are then computable from Fermi's golden rule with a contact-style interaction — the topic of §8.3.

Quick-recall checklist for §8.1–§8.2

- Three modes; their nucleon-level reactions; their atomic-mass Q-formulas; the $2m_ec^2$ threshold for β^+.
- Why EC always wins at low Q, and why it dominates at high Z.
- The continuous-spectrum puzzle as the entry point to the neutrino hypothesis.
- Three conservation laws restored by the neutrino: energy, momentum, angular momentum.
- Lepton number assignments and how they fix ν_e vs. $\bar{\nu}_e$.
- Sargent's Q^5 scaling (preview): allowed-decay phase space.

Next up (§8.3): Fermi's theory — the contact four-fermion interaction, the matrix element $|M_{fi}|^2$, the statistical (phase-space) factor, the role of the Coulomb correction $F(Z, W)$, and the Kurie plot. The spectrum shape will fall out of Fermi's golden rule in a few lines. Hold the three Q formulas in your head; they are the bridge.